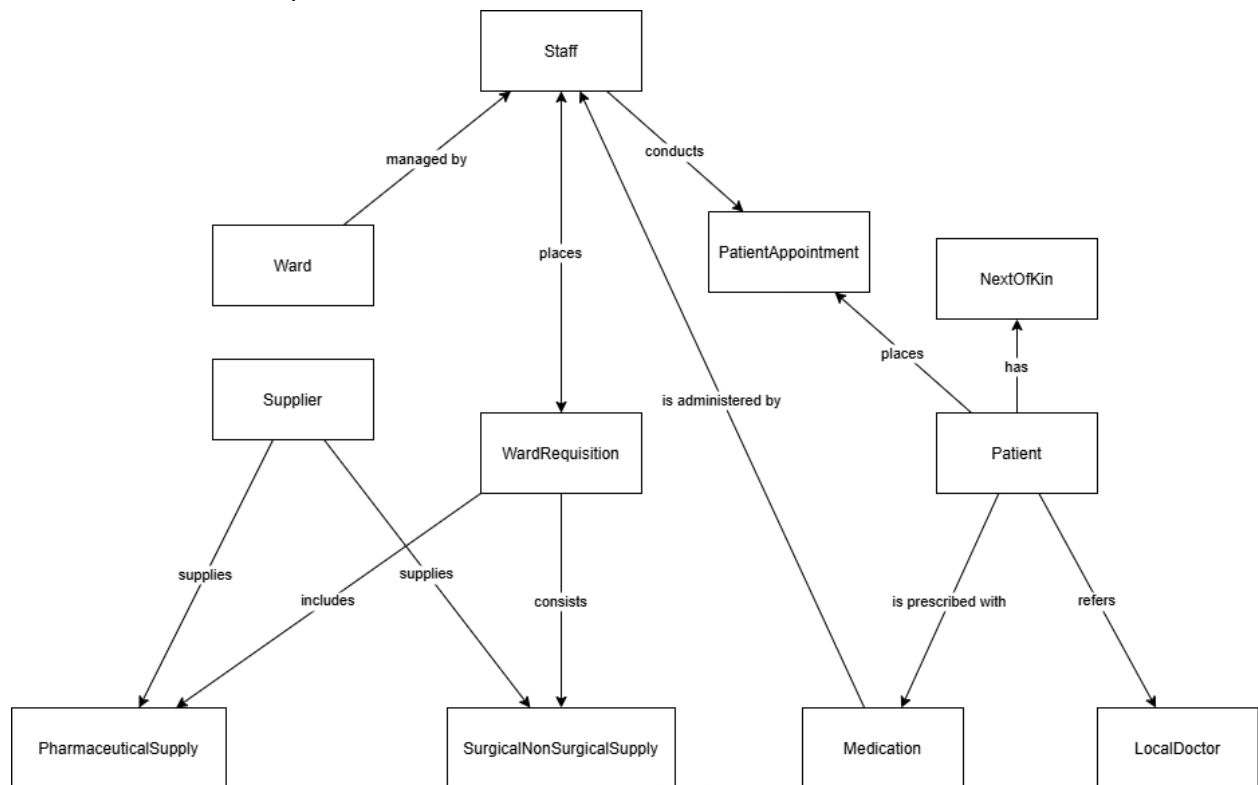


Gloren Joy Enopia Roque
BSIT BTM 2B

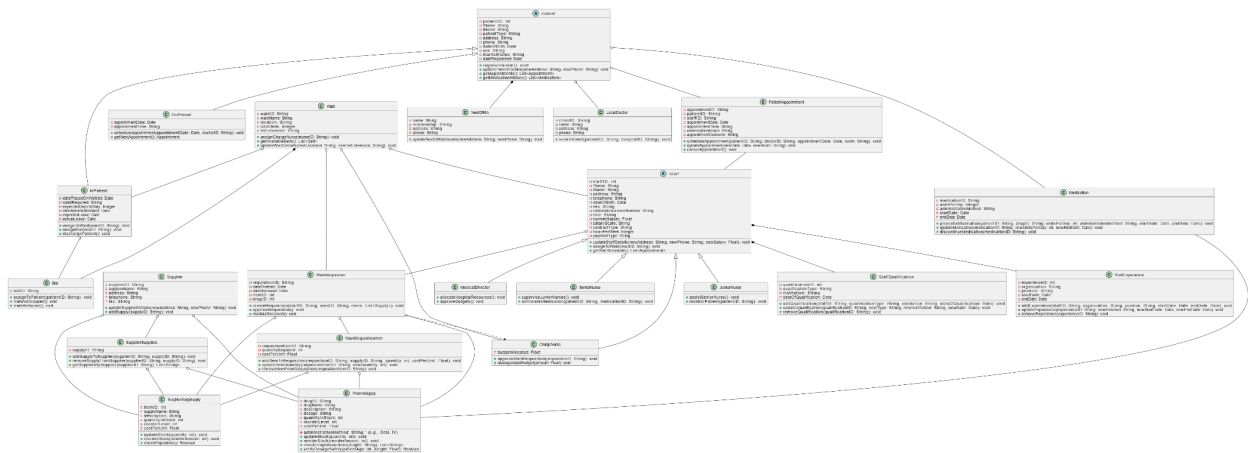
1. ERD as conceptual data model for relational DB.



link:

https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&highlight=0000ff&edit=_blank&layers=1&nav=1&title=meadows.drawio&page-id=VAWSS8wr5T6afgG8z-1P&dark=auto#Uhttps%3A%2F%2Fraw.githubusercontent.com%2Ffreynchick%2Fmeadows%2Ffreyn%2Fmeadows.drawio

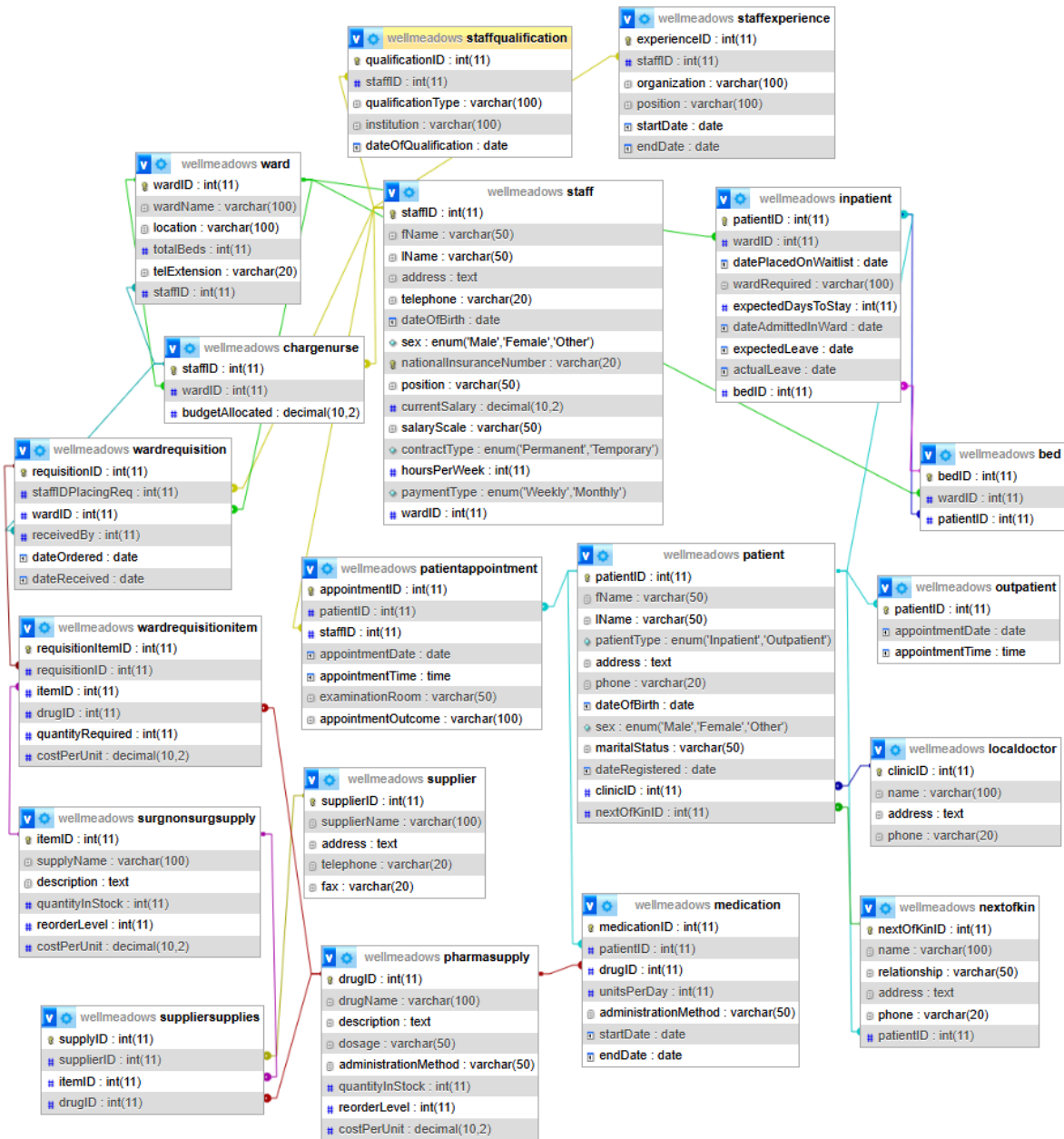
2. Class Diagram as object-oriented data model for non-relational DB.



link:

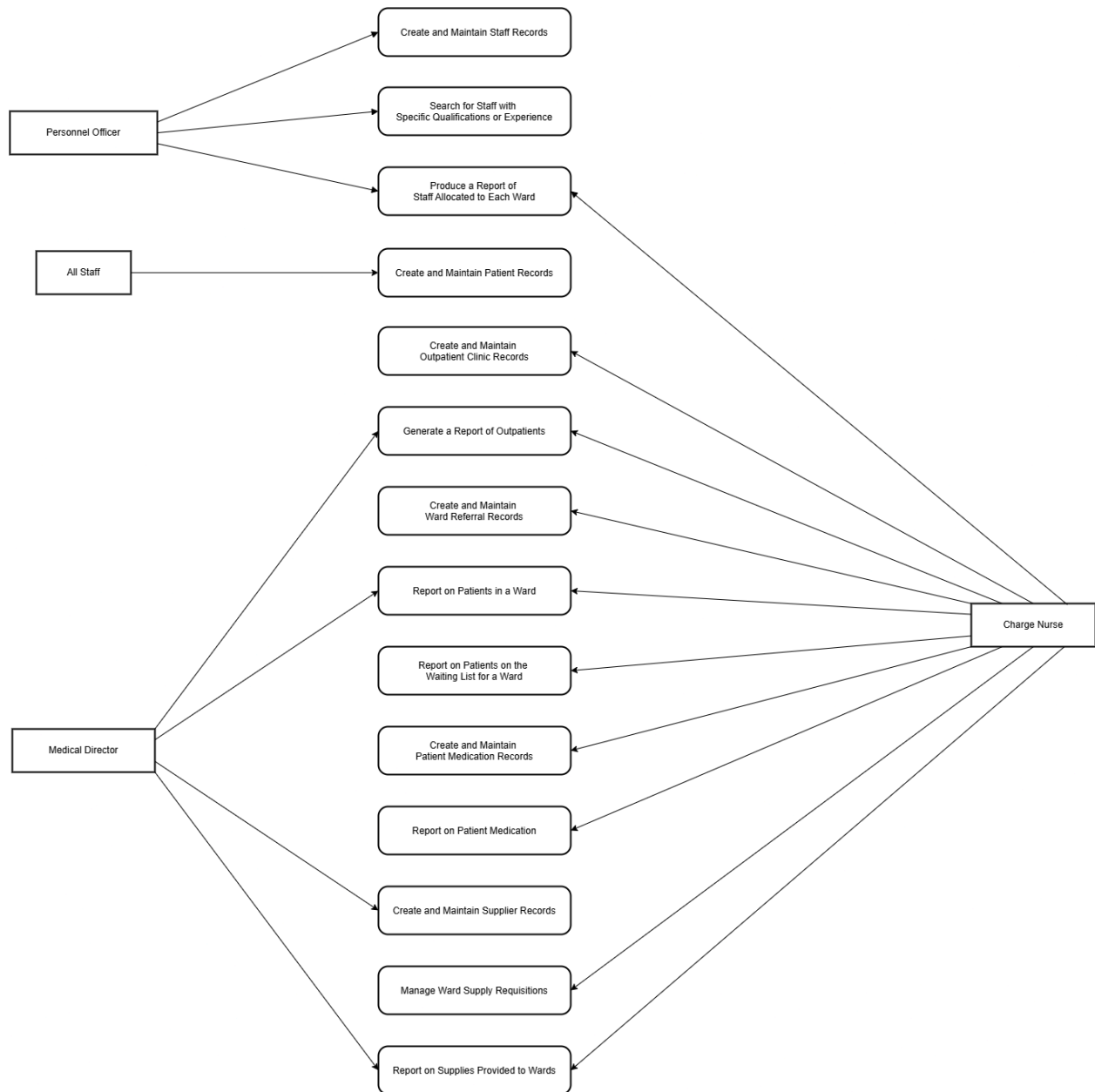
https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&highlight=0000ff&edit=_blank&layers=1&nav=1&title=meadows.drawio&dark=auto#Uhttps%3A%2F%2Fraw.githubusercontent.com%2Ffrenychick%2Fmeadows%2Ffreny%2Fmeadows.drawio

3. Implement it in a relational database and provide the physical database schema.



4. Run and test the transaction requirements and document it.

Use Cases:



```
CREATE USER 'roque_personnel_officer'@'localhost' IDENTIFIED BY '123';  
CREATE USER 'roque_charge_nurse'@'localhost' IDENTIFIED BY '123';  
CREATE USER 'roque_medical_director'@'localhost' IDENTIFIED BY '123';  
CREATE USER 'roque_all_staff'@'localhost' IDENTIFIED BY '123';
```

-- (a) Create and maintain records recording the details of members of staff (Personnel Officer)

GRANT SELECT, INSERT, UPDATE, DELETE ON staff TO
'roque_personnel_officer'@'localhost';

-- Procedure to Add a New Staff Member (INSERT)

DELIMITER \$

CREATE PROCEDURE AddStaff(

IN p_staffID INT,
IN p_fName VARCHAR(50),
IN p_lName VARCHAR(50),
IN p_address TEXT,
IN p_telephone VARCHAR(20),
IN p_dateOfBirth DATE,
IN p_sex ENUM('Male', 'Female', 'Other'),
IN p_nationalInsuranceNumber VARCHAR(20),
IN p_position VARCHAR(50),
IN p_currentSalary DECIMAL(10,2),
IN p_salaryScale VARCHAR(50),
IN p_contractType ENUM('Permanent', 'Temporary'),
IN p_hoursPerWeek INT,
IN p_paymentType ENUM('Weekly', 'Monthly'),
IN p_wardID INT

)

BEGIN

INSERT INTO staff (staffID, fName, lName, address, telephone, dateOfBirth, sex,
nationalInsuranceNumber, position, currentSalary, salaryScale, contractType, hoursPerWeek,
paymentType, wardID)

VALUES (p_staffID, p_fName, p_lName, p_address, p_telephone, p_dateOfBirth, p_sex,
p_nationalInsuranceNumber, p_position, p_currentSalary, p_salaryScale, p_contractType,
p_hoursPerWeek, p_paymentType, p_wardID);

END \$\$

-- Procedure to Update Staff Information (UPDATE)

DELIMITER \$\$

CREATE PROCEDURE UpdateStaff(

IN p_staffID INT,
IN p_address TEXT,
IN p_telephone VARCHAR(20),
IN p_currentSalary DECIMAL(10,2),
IN p_salaryScale VARCHAR(50)

)

```

BEGIN
    UPDATE staff
    SET address = p_address, telephone = p_telephone, currentSalary = p_currentSalary,
    salaryScale = p_salaryScale
    WHERE staffID = p_staffID;
END $$

```

-- Procedure to Delete a Staff Information (DELETE)

```

DELIMITER $$
CREATE PROCEDURE DeleteStaff(
    IN p_staffID INT
)
BEGIN
    DELETE FROM staff WHERE staffID = p_staffID;
END $$

```

DELIMITER ;

```

GRANT EXECUTE ON PROCEDURE AddStaff TO 'roque_personnel_officer'@'localhost';
GRANT EXECUTE ON PROCEDURE UpdateStaff TO 'roque_personnel_officer'@'localhost';
GRANT EXECUTE ON PROCEDURE DeleteStaff TO 'roque_personnel_officer'@'localhost';

```

```

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0083 seconds.)

GRANT SELECT, INSERT, UPDATE, DELETE ON staff TO 'roque_personnel_officer'@'localhost';

[ Edit inline ] [ Edit ] [ Create PHP code ]

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0120 seconds.)

CREATE PROCEDURE AddStaff( IN p_staffID INT, IN p_fName VARCHAR(50), IN p_lName VARCHAR(50), IN p_address TEXT, IN p_telephone VARCHAR(20), IN p_dateOfBirth
DATE, IN p_sex ENUM('Male', 'Female', 'Other'), IN p_nationalInsuranceNumber VARCHAR(20), IN p_position VARCHAR(50), IN p_currentSalary DECIMAL(10,2), IN
p_salaryScale VARCHAR(50), IN p_contractType ENUM('Permanent', 'Temporary'), IN p_hoursPerWeek INT, IN p_paymentType ENUM('Weekly', 'Monthly'), IN p_wardID INT
) BEGIN INSERT INTO staff (staffID, fName, lName, address, telephone, dateOfBirth, sex, nationalInsuranceNumber, position, currentSalary, salaryScale,
contractType, hoursPerWeek, paymentType, wardID) VALUES ( p_staffID, p_fName, p_lName, p_address, p_telephone, p_dateOfBirth, p_sex, p_nationalInsuranceNumber,
p_position, p_currentSalary, p_salaryScale, p_contractType, p_hoursPerWeek, p_paymentType, p_wardID); END;

[ Edit inline ] [ Edit ] [ Create PHP code ]

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0141 seconds.)

CREATE PROCEDURE UpdateStaff( IN p_staffID INT, IN p_address TEXT, IN p_telephone VARCHAR(20), IN p_currentSalary DECIMAL(10,2), IN p_salaryScale VARCHAR(50) )
BEGIN UPDATE staff SET address = p_address, telephone = p_telephone, currentSalary = p_currentSalary, salaryScale = p_salaryScale WHERE staffID = p_staffID; END;

[ Edit inline ] [ Edit ] [ Create PHP code ]

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0122 seconds.)

CREATE PROCEDURE DeleteStaff( IN p_staffID INT ) BEGIN DELETE FROM staff WHERE staffID = p_staffID; END;

[ Edit inline ] [ Edit ] [ Create PHP code ]

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0051 seconds.)

GRANT EXECUTE ON PROCEDURE AddStaff TO 'roque_personnel_officer'@'localhost';

[ Edit inline ] [ Edit ] [ Create PHP code ]

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0033 seconds.)

GRANT EXECUTE ON PROCEDURE UpdateStaff TO 'roque_personnel_officer'@'localhost';

[ Edit inline ] [ Edit ] [ Create PHP code ]

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0041 seconds.)

GRANT EXECUTE ON PROCEDURE DeleteStaff TO 'roque_personnel_officer'@'localhost';

[ Edit inline ] [ Edit ] [ Create PHP code ]

```

SHOW GRANTS FOR 'roque_personnel_officer'@'localhost';

Your SQL query has been executed successfully.

```
SHOW GRANTS FOR 'roque_personnel_officer'@'localhost'
```

☐ Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Grants for roque_personnel_officer@localhost

```
GRANT USAGE ON *.* TO 'roque_personnel_officer'@'l...
GRANT SELECT, INSERT, UPDATE, DELETE ON `wellmeado...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updatest...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addstaff...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`deletest...
```

-- Test Cases

CALL AddStaff(101, 'John', 'Doe', '123 Main St', '123-456-7890', '1990-05-15', 'Male', 'NI123456', 'Nurse', 45000.00, 'Scale A', 'Permanent', 40, 'Monthly', 1);

SELECT * FROM staff WHERE fName = 'John' AND lName = 'Doe';

CALL UpdateStaff(101, 'New Address', '555-123-4567', 48000.00, 'Scale B');

SELECT * FROM staff WHERE staffID = 101;

CALL DeleteStaff(101);

SELECT * FROM staff WHERE nationalInsuranceNumber = 'NI123456';

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0079 seconds)

```
CALL AddStaff(101, 'John', 'Doe', '123 Main St', '123-456-7890', '1990-05-15', 'Male', 'NI123456', 'Nurse', 45000.00, 'Scale A', 'Permanent', 40, 'Monthly', 1);
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 0 (1 total, Query took 0.0009 seconds)

```
SELECT * FROM staff WHERE fName = 'John' AND lName = 'Doe';
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	staffID	fName	lName	address	telephone	dateOfBirth	sex	nationalInsuranceNumber	position	currentSalary	salaryScale	contractType	hoursPerWeek	paymentType	wardID
	101	John	Doe	123 Main St	123-456-7890	1990-05-15	Male	NI123456	Nurse	45000.00	Scale A	Permanent	40	Monthly	1

☐ Check all | With selected: Edit Copy Delete Export

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0052 seconds.)

```
CALL UpdateStaff(101, 'New Address', '555-123-4567', 40000.00, 'Scale B');
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

Showing rows 0 - 0 (1 total. Query took 0.0007 seconds.)

```
SELECT * FROM staff WHERE staffID = 101;
```

☐ [Profiling](#) [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

staffID

101

fName

John

lName

Doe

address

New Address

telephone

555-123-4567

dateOfBirth

1990-05-15

sex

Male

nationalInsuranceNumber

NI123456

position

Nurse

currentSalary

48000.00

salaryScale

Scale B

contractType

Permanent

hoursPerWeek

40

paymentType

Monthly

wardID

1

☐ [Edit](#) [Copy](#) [Delete](#)

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0045 seconds.)

```
CALL DeleteStaff(101);
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0006 seconds.)

```
SELECT * FROM staff WHERE nationalInsuranceNumber = 'NI123456';
```

☐ [Profiling](#) [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

staffID

fName

lName

address

telephone

dateOfBirth

sex

nationalInsuranceNumber

position

currentSalary

salaryScale

contractType

hoursPerWeek

paymentType

wardID

Query results operations

-- (b) Search for staff who have particular qualifications or previous work experience (Personnel Officer)

```
GRANT SELECT ON wellmeadows.staffQualification TO 'roque_personnel_officer'@'localhost';
```

```
GRANT SELECT ON wellmeadows.staffExperience TO 'roque_personnel_officer'@'localhost';
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0039 seconds.)

```
-- (b) Search for staff who have particular qualifications or previous work experience (Personnel Officer) GRANT SELECT ON wellmeadows.staffQualification TO 'roque_personnel_officer'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0037 seconds.)

```
GRANT SELECT ON wellmeadows.staffExperience TO 'roque_personnel_officer'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

```
SHOW GRANTS FOR 'roque_personnel_officer'@'localhost';
```

Your SQL query has been executed successfully.

```
SHOW GRANTS FOR 'roque_personnel_officer'@'localhost'
```

Extra options

Grants for roque_personnel_officer@localhost

```
GRANT USAGE ON *.* TO 'roque_personnel_officer'@'l...
```

```
GRANT SELECT ON `wellmeadows`.`staffqualification`...
```

```
GRANT SELECT, INSERT, UPDATE, DELETE ON `wellmeado...
```

```
GRANT SELECT ON `wellmeadows`.`staffexperience` TO...
```

```
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updatest...
```

```
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addstaff...
```

```
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`deletest...
```

-- SQL query to search for staff by qualification:

```
SELECT s.staffID, s.fName, s.lName, sq.qualificationType, sq.institution, sq.dateOfQualification
FROM staff s
JOIN staffQualification sq ON s.staffID = sq.staffID
WHERE sq.qualificationType LIKE '%Nursing%';
```

✓ Showing rows 0 - 11 (12 total, Query took 0.0012 seconds.)

-- SQL query to search for staff by qualification: `SELECT s.staffID, s.fName, s.lName, sq.qualificationType, sq.institution, sq.dateOfQualification FROM staff s JOIN staffQualification sq ON s.staffID = sq.staffID WHERE sq.qualificationType LIKE '%Nursing%';`

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

staffID	fName	lName	qualificationType	institution	dateOfQualification
1	Gloren	Roque	Bachelor of Science in Nursing	University of Health Sciences	2013-05-15
5	Victoriana	Enopia	Pediatric Nursing Certification	Saint Luke Nursing School	2014-09-05
14	Sugar	Enopia	Critical Care Nursing Certification	National ICU Training Institute	2014-08-20
18	Alih	Alegre	Bachelor of Science in Nursing	University of the Philippines	2015-03-20
27	Samantha	Celeste	Pediatric Nursing Certification	Philippine Pediatric Society	2016-10-10
29	Jasmine	Dayon	Geriatric Nursing Certification	Elderly Care Academy	2014-11-12
31	Cielo	Dela Cruz	Intensive Care Nursing Certification	Philippine Nurses Association	2017-02-20
41	Carl	Mancao	Bachelor of Science in Nursing	University of the Philippines	2017-03-30
45	Gabriel	Pabillo	Bachelor of Science in Nursing	De La Salle University	2018-04-22
49	Ivan	Pavo	Bachelor of Science in Nursing	University of the Philippines	2012-05-30
53	Bralanta	Seloterio	Bachelor of Science in Nursing	Cebu Doctors' University	2015-05-10
55	Aize	Virtudazo	Bachelor of Science in Nursing	St. Luke's College of Nursing	2014-09-18

-- SQL query to search for staff by work experience:

```
SELECT s.staffID, s.fName, s.lName, se.organization, se.position, se.startDate, se.endDate
FROM staff s
JOIN staffExperience se ON s.staffID = se.staffID
WHERE se.position LIKE '%Senior Nurse%';
```

✓ Showing rows 0 - 1 (2 total, Query took 0.0005 seconds.)

```
-- SQL query to search for staff by work experience: SELECT s.staffID, s.fName, s.lName, se.organization, se.position, se.startDate, se.endDate FROM staff s JOIN staffExperience se ON s.staffID = se.staffID WHERE se.position LIKE '%Senior Nurse%';
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

staffID	fName	lName	organization	position	startDate	endDate
1	Gloren	Roque	City Hospital	Senior Nurse	2015-06-01	2023-12-31
41	Carl	Mancao	St. Luke's Medical Center	Senior Nurse	2018-03-15	2023-12-30

-- Find staff by qualification and work experience:

```
SELECT s.staffID, s.fName, s.lName, sq.qualificationType, sq.institution, se.organization, se.position
```

```
FROM staff s
```

```
LEFT JOIN staffQualification sq ON s.staffID = sq.staffID
```

```
LEFT JOIN staffExperience se ON s.staffID = se.staffID
```

```
WHERE sq.qualificationType LIKE '%Surgery%'
```

```
AND se.position LIKE '%Surgeon%';
```

✓ Showing rows 0 - 4 (5 total, Query took 0.0005 seconds.)

```
-- Find staff by qualification and work experience: SELECT s.staffID, s.fName, s.lName, sq.qualificationType, sq.institution, se.organization, se.position FROM staff s LEFT JOIN staffQualification sq ON s.staffID = sq.staffID LEFT JOIN staffExperience se ON s.staffID = se.staffID WHERE sq.qualificationType LIKE '%Surgery%' AND se.position LIKE '%Surgeon%';
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

staffID	fName	lName	qualificationType	institution	organization	position
3	Lorenzo	Roque	Master of Surgery	National Institute of Medical Studies	National Health Institute	Surgeon
7	Pablo	Roque	Orthopedic Surgery Fellowship	Global Medical Institute	Regional Medical Center	Orthopedic Surgeon
10	Lou	Enopia	Pediatric Surgery Residency	Children's Hospital Medical School	Children's Hospital	Pediatric Surgeon
15	Love	Enopia	General Surgery Board Certification	University of Surgical Sciences	Surgical Ward - Central Hospital	Surgeon
32	Louie	Delos Santos	General Surgery Residency	Philippine General Hospital	National Surgery Unit	General Surgeon

-- (c) Produce a report listing the details of staff allocated to each ward (Personnel Officer and Charge Nurse)

```
GRANT SELECT ON wellmeadows.staff TO 'roque_personnel_officer'@'localhost';
GRANT SELECT ON wellmeadows.staff TO 'roque_charge_nurse'@'localhost';
```

```
-- Procedure to get details of staffs allocated to each ward
DELIMITER $$
```

```
CREATE PROCEDURE GetWardStaffReport()
BEGIN
    SELECT s.staffID, s.fName, s.lName, s.position, w.wardID, w.wardName, w.location
    FROM staff s
    LEFT JOIN ward w ON s.wardID = w.wardID
    WHERE s.wardID IS NOT NULL;
END $$
```

```
DELIMITER ;
```

```
GRANT EXECUTE ON PROCEDURE GetWardStaffReport TO
'roque_personnel_officer'@'localhost';
GRANT EXECUTE ON PROCEDURE GetWardStaffReport TO
'roque_charge_nurse'@'localhost';
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0064 seconds.)

```
GRANT SELECT ON wellmeadows.staff TO 'roque_personnel_officer'@'localhost';
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0038 seconds.)

```
GRANT SELECT ON wellmeadows.staff TO 'roque_charge_nurse'@'localhost';
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0118 seconds.)

```
CREATE PROCEDURE GetWardStaffReport() BEGIN SELECT s.staffID, s.fName, s.lName, s.position, w.wardID, w.wardName, w.location FROM staff s LEFT JOIN
ward w ON s.wardID = w.wardID WHERE s.wardID IS NOT NULL; END;
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0033 seconds.)

```
GRANT EXECUTE ON PROCEDURE GetWardStaffReport TO 'roque_personnel_officer'@'localhost';
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0034 seconds.)

```
GRANT EXECUTE ON PROCEDURE GetWardStaffReport TO 'roque_charge_nurse'@'localhost';
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

SHOW GRANTS FOR 'roque_personnel_officer'@'localhost';

Grants for roque_personnel_officer@localhost

```
GRANT USAGE ON *.* TO 'roque_personnel_officer'@'l...
GRANT SELECT ON `wellmeadows`.`staffqualification`...
GRANT SELECT ON `wellmeadows`.`staffexperience` TO...
GRANT SELECT, INSERT, UPDATE, DELETE ON `wellmeado...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updatest...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwards...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`deletest...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addstaff...
```

SHOW GRANTS FOR 'roque_charge_nurse'@'localhost';

Grants for roque_charge_nurse@localhost

```
GRANT USAGE ON *.* TO 'roque_charge_nurse'@'localh...
GRANT SELECT ON `wellmeadows`.`staff` TO 'roque_ch...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwards...
```

-- Test Cases

CALL GetWardStaffReport()

✓ Showing rows 0 - 24 (57 total, Query took 0.0009 seconds.)

CALL GetWardStaffReport();

1 ▾

> >>

☐

Show all

Number of rows:

25 ▾

Filter rows:

Search this table

Extra options

staffID	fName	lName	position	wardID	wardName	location
1	Gloren	Roque	Charge Nurse	1	Orthopaedic	E Block
2	Grace	Roque	Charge Nurse	2	Cardiology	D Block
3	Lorenzo	Roque	Charge Nurse	3	Neurology	C Block
4	Glen	Roque	Charge Nurse	4	Pediatrics	B Block
5	Victoriana	Enopia	Charge Nurse	5	Oncology	A Block
6	Mardonio	Enopia	Charge Nurse	6	Geriatrics	F Block
7	Pablo	Roque	Charge Nurse	7	Emergency	G Block
8	Plesing	Roque	Charge Nurse	8	ICU	H Block
9	Jade	Enopia	Charge Nurse	9	Surgery	I Block
10	Lou	Enopia	Charge Nurse	10	Psychiatry	J Block
11	Lorna	Castro	Charge Nurse	11	Maternity	K Block
12	Jemmy	Boy	Charge Nurse	12	Nephrology	L Block
13	Sweet	Enopia	Charge Nurse	13	Dermatology	M Block
14	Sugar	Enopia	Charge Nurse	14	Endocrinology	N Block
15	Love	Enopia	Charge Nurse	15	Gastroenterology	O Block

-- (d) Create and maintain records recording the details of patients referred to the hospital (all staff)

```
-- Grant privileges to allow 'roque_all_staff' to manage patients
GRANT SELECT, INSERT, UPDATE ON patient TO 'roque_all_staff'@'localhost';
GRANT SELECT, INSERT, UPDATE ON inpatient TO 'roque_all_staff'@'localhost';
GRANT SELECT, INSERT, UPDATE ON outpatient TO 'roque_all_staff'@'localhost';
```

```
-- Procedure to Add a New Patient (Handles Both Inpatients & Outpatients)
DELIMITER $$
```

```
CREATE PROCEDURE AddPatient(
    IN p_patientID INT,          -- Manually provided patientID
    IN p_fName VARCHAR(50),
    IN p_lName VARCHAR(50),
    IN p_patientType ENUM('Inpatient', 'Outpatient'),
    IN p_address TEXT,
    IN p_phone VARCHAR(20),
    IN p_dateOfBirth DATE,
    IN p_sex ENUM('Male', 'Female', 'Other'),
    IN p_maritalStatus VARCHAR(50),
    IN p_dateRegistered DATE,
    IN p_clinicID INT,
    IN p_wardID INT,             -- Only for inpatients
    IN p_wardRequired VARCHAR(100), -- Only for inpatients
    IN p_expectedDaysToStay INT,  -- Only for inpatients
    IN p_dateAdmittedInWard DATE, -- Only for inpatients
    IN p_expectedLeave DATE,       -- Only for inpatients
    IN p_actualLeave DATE,         -- Only for inpatients
    IN p_bedID INT,              -- Only for inpatients
    IN p_appointmentDate DATE,    -- Only for outpatients
    IN p_appointmentTime TIME    -- Only for outpatients
)
BEGIN
    -- Step 1: Check if patientID already exists
    IF EXISTS (SELECT 1 FROM patient WHERE patientID = p_patientID) THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Error: patientID already exists';
    ELSE
        -- Step 2: Insert into patient table
        INSERT INTO patient (patientID, fName, lName, patientType, address, phone, dateOfBirth,
sex, maritalStatus, dateRegistered, clinicID)
VALUES (p_patientID, p_fName, p_lName, p_patientType, p_address, p_phone,
p_dateOfBirth, p_sex, p_maritalStatus, p_dateRegistered, p_clinicID);

        -- Step 3: If the patient is an inpatient, insert into inpatient table
        IF p_patientType = 'Inpatient' THEN
            INSERT INTO inpatient (patientID, wardID, datePlacedOnWaitlist, wardRequired,
expectedDaysToStay,
```

```

        dateAdmittedInWard, expectedLeave, actualLeave, bedID)
VALUES (p_patientID, p_wardID, CURDATE(), p_wardRequired,
p_expectedDaysToStay,
        p_dateAdmittedInWard, p_expectedLeave, p_actualLeave, p_bedID);
END IF;

```

```

-- Step 4: If the patient is an outpatient, insert into outpatient table
IF p_patientType = 'Outpatient' THEN
    INSERT INTO outpatient (patientID, appointmentDate, appointmentTime)
    VALUES (p_patientID, p_appointmentDate, p_appointmentTime);
END IF;
END IF;
END $$

```

DELIMITER ;

```

-- Procedure to Update Patient Information
DELIMITER $$

```

```

CREATE PROCEDURE UpdatePatient(
    IN p_patientID INT,
    IN p_fName VARCHAR(50),
    IN p_lName VARCHAR(50),
    IN p_patientType ENUM('Inpatient', 'Outpatient'),
    IN p_address TEXT,
    IN p_phone VARCHAR(20),
    IN p_dateOfBirth DATE,
    IN p_sex ENUM('Male', 'Female', 'Other'),
    IN p_maritalStatus VARCHAR(50),
    IN p_dateRegistered DATE,
    IN p_clinicID INT,
    IN p_nextOfKinID INT
)

```

```

BEGIN
    UPDATE patient
    SET fName = p_fName,
        lName = p_lName,
        patientType = p_patientType,
        address = p_address,
        phone = p_phone,
        dateOfBirth = p_dateOfBirth,
        sex = p_sex,
        maritalStatus = p_maritalStatus,
        dateRegistered = p_dateRegistered,
        clinicID = p_clinicID,
        nextOfKinID = p_nextOfKinID
    WHERE patientID = p_patientID;
END $$

```

DELIMITER ;

```
-- Procedure to Search for a Patient by Name
DELIMITER ;
```

```
DELIMITER $$
```

```
CREATE PROCEDURE SearchPatient(
    IN p_fName VARCHAR(50),
    IN p_lName VARCHAR(50)
)
BEGIN
    -- Select patient details
    SELECT p.*,
        (CASE
            WHEN i.patientID IS NOT NULL THEN 'Inpatient'
            WHEN o.patientID IS NOT NULL THEN 'Outpatient'
            ELSE 'Unknown'
        END) AS Patient_Status
    FROM patient p
    LEFT JOIN inpatient i ON p.patientID = i.patientID
    LEFT JOIN outpatient o ON p.patientID = o.patientID
    WHERE p.fName LIKE CONCAT('%', p_fName, '%')
        AND p.lName LIKE CONCAT('%', p_lName, '%');
END $$
```

```
DELIMITER ;
```

```
GRANT EXECUTE ON PROCEDURE AddPatient TO 'roque_all_staff'@'localhost';
GRANT EXECUTE ON PROCEDURE UpdatePatient TO 'roque_all_staff'@'localhost';
GRANT EXECUTE ON PROCEDURE SearchPatient TO 'roque_all_staff'@'localhost';
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0035 seconds.)

```
-- Grant privileges to allow 'all_staffs' to manage patients GRANT SELECT, INSERT, UPDATE ON patient TO 'roque_all_staff'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0032 seconds.)

```
GRANT SELECT, INSERT, UPDATE ON inpatient TO 'roque_all_staff'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0031 seconds.)

```
GRANT SELECT, INSERT, UPDATE ON outpatient TO 'roque_all_staff'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0143 seconds.)

```
CREATE PROCEDURE AddPatient( IN p_patientID INT, -- Manually provided patientID IN p_fName VARCHAR(50), IN p_lName VARCHAR(50), IN p_patientType ENUM('Inpatient', 'Outpatient'), IN p_address TEXT, IN
p_phone VARCHAR(20), IN p_dateOfBirth DATE, IN p_sex ENUM('Male', 'Female', 'Other'), IN p_maritalStatus VARCHAR(50), IN p_dateRegistered DATE, IN p_clinicID INT, IN p_wardID INT, -- Only for inpatients IN
p_wardRequired VARCHAR(100), -- Only for inpatients IN p_expectedDaysToStay INT, -- Only for inpatients IN p_dateAdmittedInward DATE, -- Only for inpatients IN p_expectedLeave DATE, -- Only for inpatients
IN p_actualLeave DATE, -- Only for inpatients IN p_bedID INT, -- Only for inpatients IN p_appointmentDate DATE, -- Only for outpatients IN p_appointmentTime TIME -- Only for ou[...]

```

[Edit]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0112 seconds.)

```
CREATE PROCEDURE updatePatient( IN p_patientID INT, IN p_fName VARCHAR(50), IN p_lName VARCHAR(50), IN p_patientType ENUM('Inpatient', 'Outpatient'), IN p_address TEXT, IN p_phone VARCHAR(20), IN
p_dateOfBirth DATE, IN p_sex ENUM('Male', 'Female', 'Other'), IN p_maritalStatus VARCHAR(50), IN p_dateRegistered DATE, IN p_clinicID INT, IN p_nextOfKinID INT ) BEGIN UPDATE patient SET fName = p_fName,
lName = p_lName, patientType = p_patientType, address = p_address, phone = p_phone, dateOfBirth = p_dateOfBirth, sex = p_sex, maritalStatus = p_maritalStatus, dateRegistered = p_dateRegistered, clinicID =
p_clinicID, nextOfKinID = p_nextOfKinID WHERE patientID = p_patientID; END;
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0118 seconds.)

```
CREATE PROCEDURE SearchPatient( IN p_fName VARCHAR(50), IN p_lName VARCHAR(50) ) BEGIN -- Select patient details SELECT p.*, (CASE WHEN i.patientID IS NOT NULL THEN 'Inpatient' WHEN o.patientID IS NOT NULL THEN 'Outpatient' ELSE 'unknown' END) AS Patient_Status FROM patient p LEFT JOIN inpatient i ON p.patientID = i.patientID LEFT JOIN outpatient o ON p.patientID = o.patientID WHERE p.fName LIKE CONCAT('%', p_fName, '%') AND p.lName LIKE CONCAT('%', p_lName, '%'); END;
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0033 seconds.)

```
GRANT EXECUTE ON PROCEDURE AddPatient TO 'roque_all_staff'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0031 seconds.)

```
GRANT EXECUTE ON PROCEDURE UpdatePatient TO 'roque_all_staff'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0035 seconds.)

```
GRANT EXECUTE ON PROCEDURE SearchPatient TO 'roque_all_staff'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

SHOW GRANTS FOR 'roque_all_staff'@'localhost';

Grants for roque_all_staff@localhost

```
GRANT USAGE ON *.* TO 'roque_all_staff'@'localhost'..
```

```
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`pat...
```

```
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`inp...
```

```
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`out...
```

```
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`searchpa...
```

```
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updatepa...
```

```
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addpatie...
```

-- Test Cases

CALL AddPatient(

201, 'Jane', 'Doe', 'Inpatient', '123 Maple St', '555-7890', '1995-08-20',

'Female', 'Single', CURDATE(), 3,

2, 'Cardiology', 7, '2025-03-10', '2025-03-17', NULL, 101,

NULL, NULL

);

SELECT * FROM inpatient WHERE patientID = 201;

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0143 seconds.)

```
CALL AddPatient( 201, 'Jane', 'Doe', 'Inpatient', '123 Maple St', '555-7890', '1995-08-20', 'Female', 'Single', CURDATE(), 3, 2, 'Cardiology', 7, '2025-03-10', '2025-03-17', NULL, 101, NULL, NULL );
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 0 (1 total, Query took 0.0004 seconds.)

```
SELECT * FROM inpatient WHERE patientID = 201;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	patientID	wardID	datePlacedOnWaitlist	wardRequired	expectedDaysToStay	dateAdmittedInWard	expectedLeave	actualLeave	bedID
☐ Edit Copy Delete	201	2	2025-03-12	Cardiology	7	2025-03-10	2025-03-17	NULL	101

↑ ☐ Check all | With selected: Edit Copy Delete Export

CALL AddPatient(

202, 'John', 'Smith', 'Outpatient', '456 Oak St', '555-1234', '1990-01-01',

'Male', 'Married', CURDATE(), 5,

NULL, NULL, NULL, NULL, NULL, NULL, NULL,

'2025-03-15', '10:00:00'

);

SELECT * FROM outpatient WHERE patientID = 202;

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0083 seconds.)

```
CALL AddPatient( 202, 'John', 'Smith', 'Outpatient', '456 Oak St', '555-1234', '1990-01-01', 'Male', 'Married', CURDATE(), 5, NULL, NULL, NULL, NULL, NULL, NULL, '2025-03-15', '10:00:00' );
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

```
SELECT * FROM outpatient WHERE patientID = 202;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Restore column order | Number of rows: 25 | Filter rows: Search this table

Extra options

	patientID	appointmentTime	appointmentDate
☐ Edit Copy Delete	202	10:00:00	2025-03-15

↑ ☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Restore column order | Number of rows: 25 | Filter rows: Search this table

```
CALL UpdatePatient(
    201, 'Jane', 'Smith', 'Outpatient', '456 Oak St', '555-1111', '1995-08-20',
    'Female', 'Married', CURDATE(), 4, NULL
);

SELECT * FROM patient WHERE patientID = 201;

CALL SearchPatient('Jane', 'Smith');
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0049 seconds.)

```
CALL UpdatePatient( 201, 'Jane', 'Smith', 'Outpatient', '456 Oak St', '555-1111', '1995-08-20', 'Female', 'Married', CURDATE(), 4, NULL );
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)

```
SELECT * FROM patient WHERE patientID = 201;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

	patientID	fName	lName	patientType	address	phone	dateOfBirth	sex	maritalStatus	dateRegistered	clinicID	nextOfKinID
☐ Edit Copy Delete	201	Jane	Smith	Outpatient	456 Oak St	555-1111	1995-08-20	Female	Married	2025-03-12	4	NULL

[↑](#) ☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

Showing rows 0 - 0 (1 total, Query took 0.0008 seconds.)

```
CALL SearchPatient('Jane', 'Smith');
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

patientID	fName	lName	patientType	address	phone	dateOfBirth	sex	maritalStatus	dateRegistered	clinicID	nextOfKinID	Patient_Status
201	Jane	Smith	Outpatient	456 Oak St	555-1111	1995-08-20	Female	Married	2025-03-12	4	NULL	Inpatient

-- (e) Create and maintain records recording the details of patients referred to the outpatient clinic (Charge Nurse)

GRANT SELECT, INSERT, UPDATE ON wellmeadows.outpatient TO
'roque_charge_nurse'@'localhost';

DELIMITER \$\$

CREATE PROCEDURE ManageOutpatient(

 IN p_patientID INT,
 IN p_appointmentDate DATE,
 IN p_appointmentTime TIME

)

BEGIN

 DECLARE patientExists INT;

 -- Check if the patient exists in the patient table

 SELECT COUNT(*) INTO patientExists FROM patient WHERE patientID = p_patientID AND
patientType = 'Outpatient';

 -- If patient does not exist, raise an error

 IF patientExists = 0 THEN

 SIGNAL SQLSTATE '45000'

 SET MESSAGE_TEXT = 'Patient does not exist or is not an outpatient.';

 END IF;

 -- Start transaction

 START TRANSACTION;

 -- Check if the patient already has an appointment

 IF EXISTS (SELECT * FROM outpatient WHERE patientID = p_patientID) THEN

 -- Update appointment if it already exists

 UPDATE outpatient

 SET appointmentDate = p_appointmentDate, appointmentTime = p_appointmentTime

 WHERE patientID = p_patientID;

 ELSE

 -- Insert new outpatient record if no appointment exists

 INSERT INTO outpatient (patientID, appointmentDate, appointmentTime)

 VALUES (p_patientID, p_appointmentDate, p_appointmentTime);

 END IF;

 -- Commit transaction

 COMMIT;

END \$\$

DELIMITER ;
GRANT EXECUTE ON PROCEDURE ManageOutpatient TO
'roque_charge_nurse'@'localhost';

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0034 seconds.)

```
GRANT SELECT, INSERT, UPDATE ON wellmeadows.outpatient TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0160 seconds.)

```
CREATE PROCEDURE ManageOutpatient( IN p_patientID INT, IN p_appointmentDate DATE, IN p_appointmentTime TIME ) BEGIN DECLARE patientExists INT; -- Check if the patient exists in the patient table SELECT COUNT(*) INTO patientExists FROM patient WHERE patientID = p_patientID AND patientType = 'outpatient'; -- If patient does not exist, raise an error IF patientExists = 0 THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Patient does not exist or is not an outpatient.'; END IF; -- Start transaction START TRANSACTION; -- Check if the patient already has an appointment IF EXISTS (SELECT * FROM outpatient WHERE patientID = p_patientID) THEN -- Update appointment if it already exists UPDATE outpatient SET appointmentDate = p_appointmentDate, appointmentTime = p_appointmentTime WHERE patientID = p_patientID; ELSE -- Insert new outpatient record if no appo[...]
```

[Edit]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0036 seconds.)

```
GRANT EXECUTE ON PROCEDURE ManageOutpatient TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

SHOW GRANTS FOR 'roque_charge_nurse'@'localhost';

Grants for roque_charge_nurse@localhost

GRANT USAGE ON *.* TO 'roque_charge_nurse'@'localh...

GRANT SELECT ON `wellmeadows`.`staff` TO 'roque_ch...

GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`out...

GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwards...

GRANT EXECUTE ON PROCEDURE `wellmeadows`.`manageou...

```
SELECT * FROM outpatient WHERE patientID = 2;
CALL ManageOutpatient(2, '2025-04-25', '10:10:10');
SELECT * FROM outpatient WHERE patientID = 2;
```

✓ Showing rows 0 - 0 (1 total, Query took 0.0004 seconds.)

```
SELECT * FROM outpatient WHERE patientID = 2;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Restore column order | Number of rows: 25 | Filter rows:

Extra options

	patientID	appointmentTime	appointmentDate
☐	2	10:00:00	2025-03-12

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0038 seconds.)

```
CALL ManageOutpatient(2, '2025-04-25', '10:10:10');
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

✓ Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)

```
SELECT * FROM outpatient WHERE patientID = 2;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Restore column order | Number of rows: 25 | Filter rows:

Extra options

	patientID	appointmentTime	appointmentDate
☐	2	10:10:10	2025-04-25

☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

-- (f) Produce a report listing the details of patients referred to the outpatient clinic (Charge Nurse and Medical Director)

```
GRANT SELECT ON wellmeadows.outpatient TO 'roque_charge_nurse'@'localhost';
GRANT SELECT ON wellmeadows.outpatient TO 'roque_medical_director'@'localhost';
```

--Procedure to get outpatient details

```
DELIMITER $$
```

```
CREATE PROCEDURE GetOutpatientReport()
BEGIN
    SELECT p.patientID, p.fName, p.lName, p.patientType, p.address, p.phone,
           p.dateRegistered, o.appointmentDate, o.appointmentTime
    FROM patient p
    JOIN outpatient o ON p.patientID = o.patientID;
END $$
```

```
DELIMITER ;
```

```
GRANT EXECUTE ON PROCEDURE GetOutpatientReport TO
'roque_charge_nurse'@'localhost';
GRANT EXECUTE ON PROCEDURE GetOutpatientReport TO
'roque_medical_director'@'localhost';
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0035 seconds.)

```
GRANT SELECT ON wellmeadows.outpatient TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0044 seconds.)

```
GRANT SELECT ON wellmeadows.outpatient TO 'roque_medical_director'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0134 seconds.)

```
CREATE PROCEDURE GetOutpatientReport() BEGIN SELECT p.patientID, p.fName, p.lName, p.patientType, p.address, p.phone,
p.dateRegistered, o.appointmentDate, o.appointmentTime FROM patient p JOIN outpatient o ON p.patientID = o.patientID; END;
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0031 seconds.)

```
GRANT EXECUTE ON PROCEDURE GetOutpatientReport TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0045 seconds.)

```
GRANT EXECUTE ON PROCEDURE GetOutpatientReport TO 'roque_medical_director'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. ⓘ

Your SQL query has been executed successfully.

```
SHOW GRANTS FOR 'roque_charge_nurse'@'localhost';
```

☐ Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

SHOW GRANTS FOR 'roque_charge_nurse'@'localhost';

```
Grants for roque_charge_nurse@localhost
GRANT USAGE ON *.* TO 'roque_charge_nurse'@'localh...
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`out...
GRANT SELECT ON `wellmeadows`.`staff` TO 'roque_ch...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwards...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`manageou...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getoutpa...
```

SHOW GRANTS FOR 'roque_medical_director'@'localhost';

```
Grants for roque_medical_director@localhost
GRANT USAGE ON *.* TO 'roque_medical_director'@'lo...
GRANT SELECT ON `wellmeadows`.`outpatient` TO 'roq...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getoutpa...
```

CALL GetOutpatientReport()

Showing rows 0 - 5 (6 total, Query took 0.0003 seconds.)

```
CALL GetOutpatientReport();
```

[\[Edit inline \]](#)
[\[Edit \]](#)
[\[Create PHP code \]](#)

☐ Show all
 Number of rows: 25
 Filter rows:

Extra options

patientID	fName	lName	patientType	address	phone	dateRegistered	appointmentDate	appointmentTime
2	Maria	Santos	Outpatient	22 Queen's Road, Edinburgh, Scotland, UK	+44 131 665 4567	2025-03-09	2025-04-25	10:10:10
4	Ana	Reyes	Outpatient	5 King Street, Manchester, England, UK	+44 161 432 8765	2025-03-07	2025-03-14	11:30:00
6	Isabel	Diaz	Outpatient	30 Victoria Road, Liverpool, England, UK	+44 151 321 6543	2025-03-05	2025-03-16	09:45:00
8	Catherine	Lopez	Outpatient	70 Castle Hill, Windsor, England, UK	+44 1753 567 4321	2025-03-03	2025-03-18	14:15:00
10	Samantha	Velasquez	Outpatient	15 Royal Crescent, Bath, England, UK	+44 1225 789 4567	2025-03-01	2025-03-20	16:00:00
202	John	Smith	Outpatient	456 Oak St	555-1234	2025-03-12	2025-03-15	10:00:00

-- (g) Create and maintain records recording the details of patients referred to a particular ward (Charge Nurse)

GRANT SELECT, INSERT, UPDATE ON wellmeadows.inpatient TO
'roque_charge_nurse'@'localhost';

-- Procedure to Add a New Inpatient Record(INSERT)

DELIMITER \$\$

```
CREATE PROCEDURE AddInpatient(
    IN p_patientID INT,
    IN p_wardID INT,
    IN p_datePlacedOnWaitlist DATE,
    IN p_wardRequired VARCHAR(100),
    IN p_expectedDaysToStay INT,
    IN p_dateAdmittedInWard DATE,
    IN p_expectedLeave DATE,
    IN p_actualLeave DATE,
    IN p_bedID INT
)
BEGIN
    -- Check if patient exists in the patient table and is an inpatient
    IF NOT EXISTS (SELECT 1 FROM patient WHERE patientID = p_patientID AND patientType
= 'Inpatient') THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Error: Patient does not exist or is not an Inpatient';
    END IF;

    -- Check if ward exists in the ward table
    IF NOT EXISTS (SELECT 1 FROM ward WHERE wardID = p_wardID) THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Error: Ward does not exist';
    END IF;

    -- Check if patient already has an inpatient record
    IF EXISTS (SELECT 1 FROM inpatient WHERE patientID = p_patientID) THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Error: Inpatient record already exists for this patient';
    END IF;

    -- Insert into inpatient table
    INSERT INTO inpatient (patientID, wardID, datePlacedOnWaitlist, wardRequired,
expectedDaysToStay,
        dateAdmittedInWard, expectedLeave, actualLeave, bedID)
```

```
VALUES (p_patientID, p_wardID, p_datePlacedOnWaitlist, p_wardRequired,
p_expectedDaysToStay,
p_dateAdmittedInWard, p_expectedLeave, p_actualLeave, p_bedID);
```

```
END $$
```

```
DELIMITER ;
```

```
-- Procedure to Update Inpatient Information(UPDATE)
```

```
DELIMITER $$
```

```
CREATE PROCEDURE UpdateInpatient(
```

```
IN p_patientID INT,
```

```
IN p_wardID INT,
```

```
IN p_bedID INT,
```

```
IN p_dateAdmittedInWard DATE,
```

```
IN p_expectedLeave DATE,
```

```
IN p_actualLeave DATE
```

```
)
```

```
BEGIN
```

```
UPDATE inpatient
```

```
SET wardID = p_wardID, bedID = p_bedID, dateAdmittedInWard = p_dateAdmittedInWard,
expectedLeave = p_expectedLeave, actualLeave = p_actualLeave
```

```
WHERE patientID = p_patientID;
```

```
END $$
```

```
DELIMITER ;
```

```
GRANT EXECUTE ON PROCEDURE AddInpatient TO 'roque_charge_nurse'@'localhost';
```

```
GRANT EXECUTE ON PROCEDURE UpdateInpatient TO 'roque_charge_nurse'@'localhost';
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0061 seconds.)

```
GRANT SELECT, INSERT, UPDATE ON wellmeadows.inpatient TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0159 seconds.)

```
CREATE PROCEDURE AddInpatient( IN p_patientID INT, IN p_wardID INT, IN p_datePlacedOnWaitlist DATE, IN p_wardRequired VARCHAR(100), IN p_expectedDaysToStay INT, IN
p_dateAdmittedInWard DATE, IN p_expectedLeave DATE, IN p_actualLeave DATE, IN p_bedID INT ) BEGIN -- Check if patient exists in the patient table and is an inpatient IF NOT EXISTS
(SELECT 1 FROM patient WHERE patientID = p_patientID AND patientType = 'Inpatient') THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Patient does not exist or is not an
Inpatient'; END IF; -- Check if ward exists in the ward table IF NOT EXISTS (SELECT 1 FROM ward WHERE wardID = p_wardID) THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Ward
does not exist'; END IF; -- Check if patient already has an inpatient record IF EXISTS (SELECT 1 FROM inpatient WHERE patientID = p_patientID) THEN SIGNAL SQLSTAT[...]
```

[Edit]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0166 seconds.)

```
CREATE PROCEDURE UpdateInpatient( IN p_patientID INT, IN p_wardID INT, IN p_bedID INT, IN p_dateAdmittedInWard DATE, IN p_expectedLeave DATE, IN p_actualLeave DATE ) BEGIN UPDATE
inpatient SET wardID = p_wardID, bedID = p_bedID, dateAdmittedInWard = p_dateAdmittedInWard, expectedLeave = p_expectedLeave, actualLeave = p_actualLeave WHERE patientID =
p_patientID; END;
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0036 seconds.)

```
GRANT EXECUTE ON PROCEDURE AddInpatient TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0033 seconds.)

```
GRANT EXECUTE ON PROCEDURE UpdateInpatient TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

SHOW GRANTS FOR 'roque_charge_nurse'@'localhost';

```
Grants for roque_charge_nurse@localhost
GRANT USAGE ON *.* TO 'roque_charge_nurse'@'localh...
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`inp...
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`out...
GRANT SELECT ON `wellmeadows`.`staff` TO 'roque_ch...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwards...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`manageou...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updatein...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getoutpa...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addinpat...
```

INSERT INTO patient (patientID, fName, lName, patientType, address, phone, dateOfBirth, sex, maritalStatus, dateRegistered, clinicID)
VALUES (401, 'Emily', 'Davis', 'Inpatient', '789 Birch St', '555-9876', '1990-07-21', 'Female', 'Single', CURDATE(), 3);

CALL AddInpatient(
401, 1, CURDATE(), 'Orthopaedic', 14, '2025-04-01', '2025-04-15', NULL, 2
);

SELECT * FROM inpatient WHERE patientID = 401;

CALL UpdateInpatient(
401, 1, 2, '2025-04-02', '2025-04-20', '2025-04-18'
);

SELECT * FROM inpatient WHERE patientID = 401;

✓ 1 row inserted. (Query took 0.0005 seconds.)

```
INSERT INTO patient (patientID, fName, lName, patientType, address, phone, dateOfBirth, sex, maritalStatus, dateRegistered, clinicID) VALUES (401, 'Emily', 'Davis', 'Inpatient', '789 Birch St', '555-9876', '1990-07-21', 'Female', 'Single', CURDATE(), 3);
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0043 seconds.)

```
CALL AddInpatient( 401, 1, CURDATE(), 'Orthopaedic', 14, '2025-04-01', '2025-04-15', NULL, 2 );
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 0 (1 total, Query took 0.0004 seconds.)

```
SELECT * FROM inpatient WHERE patientID = 401;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	patientID	wardID	datePlacedOnWaitlist	wardRequired	expectedDaysToStay	dateAdmittedInWard	expectedLeave	actualLeave	bedID
☐ Edit ☐ Copy ☐ Delete	401	1	2025-03-12	Orthopaedic	14	2025-04-01	2025-04-15	NULL	2

☐ Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0047 seconds.)

```
CALL UpdateInpatient( 401, 1, 2, '2025-04-02', '2025-04-20', '2025-04-18' );
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 0 (1 total. Query took 0.0004 seconds.)

```
SELECT * FROM inpatient WHERE patientID = 401;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	patientID	wardID	datePlacedOnWaitlist	wardRequired	expectedDaysToStay	dateAdmittedInWard	expectedLeave	actualLeave	bedID
☐ Edit Copy Delete	401	1	2025-03-12	Orthopaedic	14	2025-04-02	2025-04-20	2025-04-18	2

↑ ☐ Check all With selected: Edit Copy Delete Export

```
CALL AddInpatient(  
    999, 5, CURDATE(), 'Neurology', 10, '2025-03-10', '2025-03-20', NULL, 105  
);
```

Error

SQL query: [Copy](#)

```
CALL AddInpatient(  
    999, 5, CURDATE(), 'Neurology', 10, '2025-03-10', '2025-03-20', NULL, 105  
);
```

MySQL said: 

#1644 - Error: Patient does not exist or is not an Inpatient

-- (h) Produce a report listing the details of patients currently located in a particular ward (Charge Nurse and Medical Director)

-- Procedure to Get Patients in a Specific Ward

DELIMITER \$\$

CREATE PROCEDURE GetWardPatients(

IN p_wardID INT

)

BEGIN

SELECT p.patientID, p.fName, p.lName, p.patientType, w.wardName, b.bedID

FROM patient p

JOIN inpatient i ON p.patientID = i.patientID

JOIN ward w ON i.wardID = w.wardID

LEFT JOIN bed b ON i.bedID = b.bedID

WHERE w.wardID = p_wardID;

END \$\$

DELIMITER ;

GRANT EXECUTE ON PROCEDURE GetWardPatients TO 'roque_charge_nurse'@'localhost';

GRANT EXECUTE ON PROCEDURE GetWardPatients TO

'roque_medical_director'@'localhost';

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0110 seconds.)

```
CREATE PROCEDURE GetWardPatients( IN p_wardID INT ) BEGIN SELECT p.patientID, p.fName, p.lName, p.patientType, w.wardName, b.bedID FROM patient p JOIN inpatient i ON p.patientID = i.patientID JOIN ward w ON i.wardID = w.wardID LEFT JOIN bed b ON i.bedID = b.bedID WHERE w.wardID = p_wardID; END;
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0032 seconds.)

```
GRANT EXECUTE ON PROCEDURE GetWardPatients TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0028 seconds.)

```
GRANT EXECUTE ON PROCEDURE GetWardPatients TO 'roque_medical_director'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. ⓘ

Your SQL query has been executed successfully.

```
SHOW GRANTS FOR 'roque_charge_nurse'@'localhost';
```

☐ Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

SHOW GRANTS FOR 'roque_charge_nurse'@'localhost';

Grants for roque_charge_nurse@localhost

```
GRANT USAGE ON *.* TO 'roque_charge_nurse'@'localh...
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`inp...
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`out...
GRANT SELECT ON `wellmeadows`.`staff` TO 'roque_ch...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwards...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`manageou...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updatein...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwardp...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addinpat...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getoutpa...
```

SHOW GRANTS FOR 'roque_medical_director'@'localhost';

Grants for roque_medical_director@localhost

```
GRANT USAGE ON *.* TO 'roque_medical_director'@'lo...
GRANT SELECT ON `wellmeadows`.`outpatient` TO 'roq...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwardp...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getoutpa...
```

CALL GetWardPatients (1);

patientID	fName	lName	patientType	wardName	bedID
1	Juan	Cruz	Inpatient	Orthopaedic	1
401	Emily	Davis	Inpatient	Orthopaedic	2

-- (i) Produce a report listing the details of patients currently on the waiting list for a particular ward (Charge Nurse and Medical Director)

```
GRANT SELECT ON inpatient TO 'roque_charge_nurse'@'localhost';
GRANT SELECT ON inpatient TO 'roque_medical_director'@'localhost';
```

DELIMITER \$\$

-- Procedure to Get Patients on the Waiting List for a Specific Ward

```
CREATE PROCEDURE GetWaitingList(
    IN p_wardID INT
)
BEGIN
    SELECT p.patientID, p.fName, p.lName, p.patientType, i.datePlacedOnWaitlist,
i.wardRequired
    FROM inpatient i
    JOIN patient p ON i.patientID = p.patientID
    WHERE i.wardID = p_wardID AND i.dateAdmittedInWard IS NULL;
END $$
```

DELIMITER ;

```
GRANT EXECUTE ON PROCEDURE GetWaitingList TO 'roque_charge_nurse'@'localhost';
GRANT EXECUTE ON PROCEDURE GetWaitingList TO 'roque_medical_director'@'localhost';
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0048 seconds.)

```
GRANT SELECT ON inpatient TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0042 seconds.)

```
GRANT SELECT ON inpatient TO 'roque_medical_director'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0117 seconds.)

```
-- Procedure to Get Patients on the Waiting List for a Specific Ward CREATE PROCEDURE GetWaitingList( IN p_wardID INT ) BEGIN SELECT p.patientID, p.fName, p.lName, p.patientType,
i.datePlacedOnWaitlist, i.wardRequired FROM inpatient i JOIN patient p ON i.patientID = p.patientID WHERE i.wardID = p_wardID AND i.dateAdmittedInWard IS NULL; END;
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0040 seconds.)

```
GRANT EXECUTE ON PROCEDURE GetWaitingList TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0029 seconds.)

```
GRANT EXECUTE ON PROCEDURE GetWaitingList TO 'roque_medical_director'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

SHOW GRANTS FOR 'roque_charge_nurse'@'localhost';

Grants for roque_charge_nurse@localhost

```
GRANT USAGE ON *.* TO 'roque_charge_nurse'@'localh...
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`inp...
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`out...
GRANT SELECT ON `wellmeadows`.`staff` TO 'roque_ch...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwards...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`manageou...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updatein...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwardp...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addinpat...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwaiti...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getoutpa...
```

SHOW GRANTS FOR 'roque_medical_directorr'@'localhost';

Grants for roque_medical_director@localhost

```
GRANT USAGE ON *.* TO 'roque_medical_director'@'lo...
GRANT SELECT ON `wellmeadows`.`outpatient` TO 'roq...
GRANT SELECT ON `wellmeadows`.`inpatient` TO 'roqu...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwardp...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getoutpa...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwaiti...
```

```
INSERT INTO patient (patientID, fName, lName, patientType, address, phone, dateOfBirth, sex, maritalStatus, dateRegistered, clinicID)
VALUES
(601, 'Alice', 'Johnson', 'Inpatient', '123 Elm St', '555-1212', '1992-06-15', 'Female', 'Single',
CURDATE(), 2),
(602, 'Bob', 'Williams', 'Inpatient', '456 Pine St', '555-2323', '1988-09-20', 'Male', 'Married',
CURDATE(), 3),
(603, 'Charlie', 'Brown', 'Inpatient', '789 Oak St', '555-3434', '1995-02-10', 'Male', 'Single',
CURDATE(), 4);
```

```
INSERT INTO inpatient (patientID, wardID, datePlacedOnWaitlist, wardRequired,
expectedDaysToStay, dateAdmittedInWard, expectedLeave, actualLeave, bedID)
VALUES
(601, 1, CURDATE(), 'Orthopaedic', 7, NULL, NULL, NULL, NULL),
(602, 1, CURDATE(), 'Orthopaedic', 10, NULL, NULL, NULL, NULL);
```

```
CALL GetWaitingList(1);
```

3 rows inserted. (Query took 0.0062 seconds.)

```
INSERT INTO patient (patientID, fName, lName, patientType, address, phone, dateOfBirth, sex, maritalStatus, dateRegistered, clinicID) VALUES (601, 'Alice', 'Johnson', 'Inpatient', '123 Elm St', '555-1212', '1992-06-15', 'Female', 'Single', CURDATE(), 2), (602, 'Bob', 'Williams', 'Inpatient', '456 Pine St', '555-2323', '1988-09-20', 'Male', 'Married', CURDATE(), 3), (603, 'Charlie', 'Brown', 'Inpatient', '789 Oak St', '555-3434', '1995-02-10', 'Male', 'Single', CURDATE(), 4);
```

[Edit inline][Edit][Create PHP code]

2 rows inserted. (Query took 0.0045 seconds.)

```
INSERT INTO inpatient (patientID, wardID, datePlacedOnWaitlist, wardRequired, expectedDaysToStay, dateAdmittedInWard, expectedLeave, actualLeave, bedID) VALUES (601, 1, CURDATE(), 'Orthopaedic', 7, NULL, NULL, NULL, NULL), (602, 1, CURDATE(), 'Orthopaedic', 10, NULL, NULL, NULL, NULL);
```

[Edit inline][Edit][Create PHP code]

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 1 (2 total, Query took 0.0006 seconds.)

```
CALL GetWaitingList(1);
```

[Edit inline][Edit][Create PHP code]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

patientID	fName	lName	patientType	datePlacedOnWaitlist	wardRequired
601	Alice	Johnson	Inpatient	2025-03-12	Orthopaedic
602	Bob	Williams	Inpatient	2025-03-12	Orthopaedic

-- (j) Create and maintain records recording the details of medication given to a particular patient (Charge Nurse)

```
GRANT SELECT, INSERT, UPDATE, DELETE ON wellmeadows.medication TO  
'roque_charge_nurse'@'localhost';
```

-- Procedure to Add Medication Record for a Patient

```
DROP PROCEDURE IF EXISTS AddMedication;  
DROP PROCEDURE IF EXISTS UpdateMedication;  
DROP PROCEDURE IF EXISTS DeleteMedication;
```

```
DELIMITER $$
```

-- Procedure to Add Medication Record

```
CREATE PROCEDURE AddMedication(  
    IN p_medicationID INT,  
    IN p_patientID INT,  
    IN p_drugID INT,  
    IN p_unitsPerDay INT,  
    IN p_administrationMethod VARCHAR(50),  
    IN p_startDate DATE,  
    IN p_endDate DATE  
)  
BEGIN  
    INSERT INTO medication (medicationID, patientID, drugID, unitsPerDay,  
administrationMethod, startDate, endDate)  
    VALUES (p_medicationID, p_patientID, p_drugID, p_unitsPerDay, p_administrationMethod,  
p_startDate, p_endDate);  
END $$
```

-- Procedure to Update Medication Record

```
CREATE PROCEDURE UpdateMedication(  
    IN p_medicationID INT,  
    IN p_patientID INT,  
    IN p_drugID INT,  
    IN p_unitsPerDay INT,  
    IN p_administrationMethod VARCHAR(50),  
    IN p_startDate DATE,  
    IN p_endDate DATE  
)  
BEGIN  
    UPDATE medication  
    SET patientID = p_patientID,  
        drugID = p_drugID,
```

```

        unitsPerDay = p_unitsPerDay,
        administrationMethod = p_administrationMethod,
        startDate = p_startDate,
        endDate = p_endDate
    WHERE medicationID = p_medicationID;
END $$

```

```

-- Procedure to Delete Medication Record
CREATE PROCEDURE DeleteMedication(
    IN p_medicationID INT
)
BEGIN
    DELETE FROM medication WHERE medicationID = p_medicationID;
END $$

```

```

DELIMITER ;

```

```

GRANT EXECUTE ON PROCEDURE AddMedication TO 'roque_charge_nurse'@'localhost';
GRANT EXECUTE ON PROCEDURE UpdateMedication TO 'roque_charge_nurse'@'localhost';
GRANT EXECUTE ON PROCEDURE DeleteMedication TO 'roque_charge_nurse'@'localhost';

```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0035 seconds.)

```
GRANT SELECT, INSERT, UPDATE, DELETE ON wellmeadows.medication TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0002 seconds.)

```
-- Procedure to Add Medication Record for a Patient DROP PROCEDURE IF EXISTS AddMedication;
```

[Edit inline] [Edit] [Create PHP code]

⚠ Note: #1305 PROCEDURE wellmeadows.AddMedication does not exist

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

```
DROP PROCEDURE IF EXISTS UpdateMedication;
```

[Edit inline] [Edit] [Create PHP code]

⚠ Note: #1305 PROCEDURE wellmeadows.UpdateMedication does not exist

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

```
DROP PROCEDURE IF EXISTS DeleteMedication;
```

[Edit inline] [Edit] [Create PHP code]

⚠ Note: #1305 PROCEDURE wellmeadows.DeleteMedication does not exist

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0201 seconds.)

```
-- Procedure to Add Medication Record CREATE PROCEDURE AddMedication( IN p_medicationID INT, IN p_patientID INT, IN p_drugID INT, IN p_unitsPerDay INT, IN p_administrationMethod
VARCHAR(50), IN p_startDate DATE, IN p_endDate DATE ) BEGIN INSERT INTO medication (medicationID, patientID, drugID, unitsPerDay, administrationMethod, startDate, endDate) VALUES
(p_medicationID, p_patientID, p_drugID, p_unitsPerDay, p_administrationMethod, p_startDate, p_endDate); END;
```

[Edit inline] [Edit] [Create PHP code]

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0135 seconds.)

```
-- Procedure to Update Medication Record CREATE PROCEDURE UpdateMedication( IN p_medicationID INT, IN p_patientID INT, IN p_drugID INT, IN p_unitsPerDay INT, IN p_administrationMethod VARCHAR(50), IN p_startDate DATE, IN p_endDate DATE ) BEGIN UPDATE medication SET patientID = p_patientID, drugID = p_drugID, unitsPerDay = p_unitsPerDay, administrationMethod = p_administrationMethod, startDate = p_startDate, endDate = p_endDate WHERE medicationID = p_medicationID; END;
```

[Edit inline] [Edit] [Create PHP code]

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0142 seconds.)

```
-- Procedure to Delete Medication Record CREATE PROCEDURE DeleteMedication( IN p_medicationID INT ) BEGIN DELETE FROM medication WHERE medicationID = p_medicationID; END;
```

[Edit inline] [Edit] [Create PHP code]

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0038 seconds.)

```
GRANT EXECUTE ON PROCEDURE AddMedication TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0070 seconds.)

```
GRANT EXECUTE ON PROCEDURE UpdateMedication TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0033 seconds.)

```
GRANT EXECUTE ON PROCEDURE DeleteMedication TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

SHOW GRANTS FOR 'roque_charge_nurse'@'localhost';

Grants for roque_charge_nurse@localhost

GRANT USAGE ON *.* TO 'roque_charge_nurse'@'localh...

GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`inp...

GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`out...

GRANT SELECT ON `wellmeadows`.`staff` TO 'roque_ch...

GRANT SELECT, INSERT, UPDATE, DELETE ON `wellmeado...

GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addmedic...

GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwards...

GRANT EXECUTE ON PROCEDURE `wellmeadows`.`manageou...

GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updatein...

GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwardp...

GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addinpat...

GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwaiti...

GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getoutpa...

GRANT EXECUTE ON PROCEDURE `wellmeadows`.`deleteme...

GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updateme...

-- Insert a Sample Patient

```
INSERT INTO patient (patientID, fName, lName, patientType, address, phone, dateOfBirth, sex, maritalStatus, dateRegistered, clinicID)
```

```
VALUES (701, 'David', 'Miller', 'Inpatient', '123 Aspen St', '555-9191', '1990-12-01', 'Male', 'Married', CURDATE(), 2);
```

-- Insert a Sample Drug into pharmaSupply (Replacing the old drug table)

```
INSERT INTO pharmaSupply (drugID, drugName, description, dosage, administrationMethod, quantityInStock, reorderLevel, costPerUnit)
```

```
VALUES (301, 'Paracetamol', 'Pain Reliever', '500mg', 'Oral', 100, 10, 5.00);
```

-- Add Medication for the Patient (Ensure drugID exists in pharmaSupply)

```
CALL AddMedication(
    1001, 701, 301, 2, 'Oral', '2025-03-15', '2025-03-25'
);
```

-- Verify that the medication was added correctly

```
SELECT * FROM medication WHERE medicationID = 1001;
```

-- Update Medication Record (Ensure `drugID` still refers to pharmaSupply)

```
CALL UpdateMedication(
    1001, 701, 301, 3, 'Intravenous', '2025-03-16', '2025-03-30'
);
```

-- Verify the Update

```
SELECT * FROM medication WHERE medicationID = 1001;
```

-- Delete Medication Record

```
CALL DeleteMedication(1001);
```

-- Verify Deletion (Should return no rows)

```
SELECT * FROM medication WHERE medicationID = 1001;
```

✔ 1 row inserted. (Query took 0.0058 seconds.)

```
-- Insert a Sample Patient INSERT INTO patient (patientID, fName, lName, patientType, address, phone, dateOfBirth, sex, maritalStatus, dateRegistered, clinicID) VALUES (701, 'David', 'Miller', 'Inpatient', '123 Aspen St', '555-9191', '1990-12-01', 'Male', 'Married', CURDATE(), 2);
```

[Edit inline] [Edit] [Create PHP code]

✔ 1 row inserted. (Query took 0.0003 seconds.)

```
-- Insert a Sample Drug into pharmaSupply (Replacing the old drug table) INSERT INTO pharmaSupply (drugID, drugName, description, dosage, administrationMethod, quantityInStock, reorderLevel, costPerUnit) VALUES (301, 'Paracetamol', 'Pain Reliever', '500mg', 'Oral', 100, 10, 5.00);
```

[Edit inline] [Edit] [Create PHP code]

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0041 seconds.)

```
-- Add Medication for the Patient (Ensure drugID exists in pharmaSupply) CALL AddMedication( 1001, 701, 301, 2, 'Oral', '2025-03-15', '2025-03-25' );
```

[Edit inline] [Edit] [Create PHP code]

✔ Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

```
-- Verify that the medication was added correctly SELECT * FROM medication WHERE medicationID = 1001;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

		medicationID	patientID	drugID	unitsPerDay	administrationMethod	startDate	endDate
☐	Edit	Copy	Delete	1001	701	301	2 Oral	2025-03-15 2025-03-25

☐ Check all | With selected: Edit Copy Delete Export

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0043 seconds.)

```
-- Update Medication Record (Ensure 'drugID' still refers to pharmaSupply) CALL UpdateMedication( 1001, 701, 301, 3, 'Intravenous', '2025-03-16', '2025-03-30' );
```

[Edit inline] [Edit] [Create PHP code]

✔ Showing rows 0 - 0 (1 total, Query took 0.0004 seconds.)

```
-- Verify the Update SELECT * FROM medication WHERE medicationID = 1001;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows:

Extra options

		medicationID	patientID	drugID	unitsPerDay	administrationMethod	startDate	endDate
☐	Edit	Copy	Delete	1001	701	301	3 Intravenous	2025-03-16 2025-03-30

☐ Check all | With selected: Edit Copy Delete Export

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0063 seconds.)

```
-- Delete Medication Record CALL DeleteMedication(1001);
```

[Edit inline] [Edit] [Create PHP code]

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0045 seconds.)

```
-- Verify Deletion (Should return no rows) SELECT * FROM medication WHERE medicationID = 1001;
```

[Edit inline] [Edit] [Create PHP code]

medicationID	patientID	drugID	unitsPerDay	administrationMethod	startDate	endDate
--------------	-----------	--------	-------------	----------------------	-----------	---------

Query results operations

-- (k) Produce a report listing the details of medication for a particular patient (Charge Nurse)

```
GRANT SELECT ON wellmeadows.medication TO 'roque_charge_nurse'@'localhost';
```

-- Procedure to Get Medication Details for a Specific Patient

```
DELIMITER $$
CREATE PROCEDURE GetPatientMedication(
  IN p_patientID INT
)
BEGIN
  SELECT m.medicationID, m.patientID, p.fName, p.lName, ps.drugName, m.unitsPerDay,
m.administrationMethod, m.startDate, m.endDate
  FROM medication m
  JOIN patient p ON m.patientID = p.patientID
  JOIN pharmaSupply ps ON m.drugID = ps.drugID
  WHERE m.patientID = p_patientID;
END $$
```

```
DELIMITER ;
```

```
GRANT EXECUTE ON PROCEDURE GetPatientMedication TO
'roque_charge_nurse'@'localhost';
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0095 seconds.)

```
GRANT SELECT ON wellmeadows.medication TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0120 seconds.)

```
CREATE PROCEDURE GetPatientMedication( IN p_patientID INT ) BEGIN SELECT m.medicationID, m.patientID, p.fName, p.lName, ps.drugName, m.unitsPerDay,
m.administrationMethod, m.startDate, m.endDate FROM medication m JOIN patient p ON m.patientID = p.patientID JOIN pharmaSupply ps ON m.drugID =
ps.drugID WHERE m.patientID = p_patientID; END;
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0054 seconds.)

```
GRANT EXECUTE ON PROCEDURE GetPatientMedication TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

SHOW GRANTS FOR 'roque_charge_nurse'@'localhost';

```
Grants for roque_charge_nurse@localhost
GRANT USAGE ON *.* TO 'roque_charge_nurse'@'localh...
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`inp...
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`out...
GRANT SELECT ON `wellmeadows`.`staff` TO 'roque_ch...
GRANT SELECT, INSERT, UPDATE, DELETE ON `wellmeado...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addmedic...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwaiti...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`manageou...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updatein...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwardp...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addinpat...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getpatie...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getoutpa...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`deleteme...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updateme...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwards...
```

CALL GetPatientMedication (2)

Showing rows 0 - 0 (1 total, Query took 0.0023 seconds.)

```
CALL GetPatientMedication (2);
```

[\[Edit inline \]](#)
[\[Edit \]](#)
[\[Create PHP code \]](#)

☐ Show all
 Number of rows: 25
 Filter rows: Search this table

Extra options

medicationID	patientID	fName	lName	drugName	unitsPerDay	administrationMethod	startDate	endDate
2	2	Maria	Santos	Amoxicillin	2	Oral	2025-03-09	2025-03-14

**-- (I) Create and maintain records recording the details of suppliers for the hospital
(Medical Director)**

```
GRANT SELECT, INSERT, UPDATE, DELETE ON wellmeadows.supplier TO  
'roque_medical_director'@'localhost';
```

```
DELIMITER $$
```

-- Procedure to Add a New Supplier

```
CREATE PROCEDURE AddSupplier(  
    IN p_supplierID INT,  
    IN p_supplierName VARCHAR(100),  
    IN p_address TEXT,  
    IN p_telephone VARCHAR(20),  
    IN p_fax VARCHAR(20)  
)  
BEGIN  
    INSERT INTO supplier (supplierID, supplierName, address, telephone, fax)  
    VALUES (p_supplierID, p_supplierName, p_address, p_telephone, p_fax);  
END $$
```

-- Procedure to Update Supplier Information

```
CREATE PROCEDURE UpdateSupplier(  
    IN p_supplierID INT,  
    IN p_supplierName VARCHAR(100),  
    IN p_address TEXT,  
    IN p_telephone VARCHAR(20),  
    IN p_fax VARCHAR(20)  
)  
BEGIN  
    UPDATE supplier  
    SET supplierName = p_supplierName, address = p_address, telephone = p_telephone, fax =  
    p_fax  
    WHERE supplierID = p_supplierID;  
END $$
```

-- Procedure to Delete a Supplier Record

```
CREATE PROCEDURE DeleteSupplier(  
    IN p_supplierID INT  
)  
BEGIN  
    DELETE FROM supplier WHERE supplierID = p_supplierID;  
END $$
```

```
DELIMITER ;
```

```
GRANT EXECUTE ON PROCEDURE AddSupplier TO 'roque_medical_director'@'localhost';
GRANT EXECUTE ON PROCEDURE UpdateSupplier TO 'roque_medical_director'@'localhost';
GRANT EXECUTE ON PROCEDURE DeleteSupplier TO 'roque_medical_director'@'localhost';
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0037 seconds.)

```
GRANT SELECT, INSERT, UPDATE, DELETE ON wellmeadows.supplier TO 'roque_medical_director'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0155 seconds.)

```
-- Procedure to Add a New Supplier CREATE PROCEDURE AddSupplier( IN p_supplierID INT, IN p_supplierName VARCHAR(100), IN p_address TEXT, IN p_telephone
VARCHAR(20), IN p_fax VARCHAR(20) ) BEGIN INSERT INTO supplier (supplierID, supplierName, address, telephone, fax) VALUES (p_supplierID,
p_supplierName, p_address, p_telephone, p_fax); END;
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0143 seconds.)

```
-- Procedure to Update Supplier Information CREATE PROCEDURE UpdateSupplier( IN p_supplierID INT, IN p_supplierName VARCHAR(100), IN p_address TEXT, IN
p_telephone VARCHAR(20), IN p_fax VARCHAR(20) ) BEGIN UPDATE supplier SET supplierName = p_supplierName, address = p_address, telephone = p_telephone,
fax = p_fax WHERE supplierID = p_supplierID; END;
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0115 seconds.)

```
-- Procedure to Delete a Supplier Record CREATE PROCEDURE DeleteSupplier( IN p_supplierID INT ) BEGIN DELETE FROM supplier WHERE supplierID =
p_supplierID; END;
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0035 seconds.)

```
GRANT EXECUTE ON PROCEDURE AddSupplier TO 'roque_medical_director'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0033 seconds.)

```
GRANT EXECUTE ON PROCEDURE UpdateSupplier TO 'roque_medical_director'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0032 seconds.)

```
GRANT EXECUTE ON PROCEDURE DeleteSupplier TO 'roque_medical_director'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

SHOW GRANTS FOR 'roque_medical_director'@'localhost';

Grants for roque_medical_director@localhost

```
GRANT USAGE ON *.* TO 'roque_medical_director'@'lo...
GRANT SELECT, INSERT, UPDATE, DELETE ON `wellmeado...
GRANT SELECT ON `wellmeadows`.`outpatient` TO `roq...
GRANT SELECT ON `wellmeadows`.`inpatient` TO `roqu...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updatesu...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getoutpa...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`deletesu...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwaiti...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addsuppl...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwardp...
```

```
CALL AddSupplier(21, 'Aegis Pharma Ltd', '100 Med St, London', '07555555555',  
'02012345678');
```

```
SELECT * FROM supplier WHERE supplierID = 21;
```

```
CALL UpdateSupplier(21, 'Aegis Pharma Ltd', '150 Med St, London', '07555555555',  
'02012345679');
```

```
SELECT * FROM supplier WHERE supplierID = 21;
```

```
CALL DeleteSupplier(21);
```

```
SELECT * FROM supplier WHERE supplierID = 21;
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0088 seconds.)

```
CALL AddSupplier(21, 'Aegis Pharma Ltd', '100 Med St, London', '07555555555', '02012345678');
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

```
SELECT * FROM supplier WHERE supplierID = 21;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	supplierID	supplierName	address	telephone	fax
☐ Edit ☐ Copy ☐ Delete	21	Aegis Pharma Ltd	100 Med St, London	07555555555	02012345678

↑ ☐ Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0047 seconds.)

```
CALL UpdateSupplier(21, 'Aegis Pharma Ltd', '150 Med St, London', '07555555555', '02012345679');
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)

```
SELECT * FROM supplier WHERE supplierID = 21;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	supplierID	supplierName	address	telephone	fax
☐ Edit ☐ Copy ☐ Delete	21	Aegis Pharma Ltd	150 Med St, London	07555555555	02012345679

↑ ☐ Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0055 seconds.)

```
CALL DeleteSupplier(21);
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
SELECT * FROM supplier WHERE supplierID = 21;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

supplierID	supplierName	address	telephone	fax
------------	--------------	---------	-----------	-----

Query results operations

**-- (m) Create and maintain records detailing requisitions for supplies for particular wards
(Charge Nurse)**

```
GRANT SELECT, INSERT, UPDATE, DELETE ON wellmeadows.wardRequisition TO  
'roque_charge_nurse'@'localhost';
```

```
DELIMITER $$
```

```
CREATE PROCEDURE AddWardRequisition(  
    IN p_requisitionID INT,  
    IN p_staffIDPlacingReq INT,  
    IN p_wardID INT,  
    IN p_receivedBy INT,  
    IN p_dateOrdered DATE,  
    IN p_dateReceived DATE  
)  
BEGIN  
    -- Ensure foreign key constraints are met  
    IF NOT EXISTS (SELECT 1 FROM staff WHERE staffID = p_staffIDPlacingReq) THEN  
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Staff placing request does not  
exist.';  
    END IF;  
    IF NOT EXISTS (SELECT 1 FROM ward WHERE wardID = p_wardID) THEN  
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Ward does not exist.';  
    END IF;  
    IF p_receivedBy IS NOT NULL AND NOT EXISTS (SELECT 1 FROM chargeNurse WHERE  
staffID = p_receivedBy) THEN  
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Receiving charge nurse does  
not exist.';  
    END IF;  
    IF p_dateReceived IS NOT NULL AND p_dateReceived < p_dateOrdered THEN  
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: dateReceived cannot be  
before dateOrdered.';  
    END IF;  
  
    -- Ensure requisitionID does not already exist  
    IF EXISTS (SELECT 1 FROM wardRequisition WHERE requisitionID = p_requisitionID)  
THEN  
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: requisitionID already exists.';  
    END IF;  
  
    -- Insert into wardRequisition with manual requisitionID  
    INSERT INTO wardRequisition (requisitionID, staffIDPlacingReq, wardID, receivedBy,  
dateOrdered, dateReceived)
```

```
VALUES (p_requisitionID, p_staffIDPlacingReq, p_wardID, p_receivedBy, p_dateOrdered,  
p_dateReceived);  
END $$
```

-- Procedure to Update a Ward Requisition

```
CREATE PROCEDURE UpdateWardRequisition(  
    IN p_requisitionID INT,  
    IN p_receivedBy INT,  
    IN p_dateReceived DATE  
)  
BEGIN  
    -- Validate requisition exists  
    IF NOT EXISTS (SELECT 1 FROM wardRequisition WHERE requisitionID = p_requisitionID)  
    THEN  
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Requisition does not exist.';  
    END IF;  
  
    -- Validate receivedBy exists in chargeNurse  
    IF NOT EXISTS (SELECT 1 FROM chargeNurse WHERE staffID = p_receivedBy) THEN  
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Receiving charge nurse does  
not exist.';  
    END IF;  
  
    -- Ensure dateReceived is not before dateOrdered  
    IF (SELECT dateOrdered FROM wardRequisition WHERE requisitionID = p_requisitionID) >  
p_dateReceived THEN  
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: dateReceived cannot be  
before dateOrdered.';  
    END IF;  
  
    -- Update requisition  
    UPDATE wardRequisition  
    SET receivedBy = p_receivedBy, dateReceived = p_dateReceived  
    WHERE requisitionID = p_requisitionID;  
END $$
```

-- Procedure to Delete a Ward Requisition

```
CREATE PROCEDURE DeleteWardRequisition(  
    IN p_requisitionID INT  
)  
BEGIN  
    -- Validate requisition exists before deleting  
    IF NOT EXISTS (SELECT 1 FROM wardRequisition WHERE requisitionID = p_requisitionID)  
    THEN
```

```
SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Requisition does not exist.';
END IF;
```

```
DELETE FROM wardRequisition WHERE requisitionID = p_requisitionID;
END $$
```

```
DELIMITER ;
```

```
GRANT EXECUTE ON PROCEDURE AddWardRequisition TO
'roque_charge_nurse'@'localhost';
GRANT EXECUTE ON PROCEDURE UpdateWardRequisition TO
'roque_charge_nurse'@'localhost';
GRANT EXECUTE ON PROCEDURE DeleteWardRequisition TO
'roque_charge_nurse'@'localhost';
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0037 seconds.)

```
GRANT SELECT, INSERT, UPDATE, DELETE ON wellmeadows.wardRequisition TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0145 seconds.)

```
CREATE PROCEDURE AddWardRequisition( IN p_requisitionID INT, IN p_staffIDPlacingReq INT, IN p_wardID INT, IN p_receivedBy INT, IN p_dateOrdered DATE, IN p_dateReceived DATE ) BEGIN -- Ensure foreign key constraints are met
IF NOT EXISTS (SELECT 1 FROM staff WHERE staffID = p_staffIDPlacingReq) THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Staff placing request does not exist.'; END IF; IF NOT EXISTS (SELECT 1 FROM ward WHERE wardID
= p_wardID) THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: ward does not exist.'; END IF; IF p_receivedBy IS NOT NULL AND NOT EXISTS (SELECT 1 FROM chargeNurse WHERE staffID = p_receivedBy) THEN SIGNAL SQLSTATE
'45000' SET MESSAGE_TEXT = 'Error: Receiving charge nurse does not exist.'; END IF; IF p_dateReceived IS NOT NULL AND p_dateReceived < p_dateOrdered THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: dateReceived
cannot be b[...]
```

[Edit]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0115 seconds.)

```
-- Procedure to Update a Ward Requisition CREATE PROCEDURE UpdateWardRequisition( IN p_requisitionID INT, IN p_receivedBy INT, IN p_dateReceived DATE ) BEGIN -- Validate requisition exists IF NOT EXISTS (SELECT 1 FROM
wardRequisition WHERE requisitionID = p_requisitionID) THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Requisition does not exist.'; END IF; -- Validate receivedBy exists in chargeNurse IF NOT EXISTS (SELECT 1 FROM
chargeNurse WHERE staffID = p_receivedBy) THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Receiving charge nurse does not exist.'; END IF; -- Ensure dateReceived is not before dateOrdered IF (SELECT dateOrdered FROM
wardRequisition WHERE requisitionID = p_requisitionID) > p_dateReceived THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: dateReceived cannot be before dateOrdered.'; END IF; -- Update requisition UPDATE
wardRequisition [...]
```

[Edit]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0133 seconds.)

```
-- Procedure to Delete a Ward Requisition CREATE PROCEDURE DeleteWardRequisition( IN p_requisitionID INT ) BEGIN -- Validate requisition exists before deleting IF NOT EXISTS (SELECT 1 FROM wardRequisition WHERE
requisitionID = p_requisitionID) THEN SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Error: Requisition does not exist.'; END IF; DELETE FROM wardRequisition WHERE requisitionID = p_requisitionID; END;
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0038 seconds.)

```
GRANT EXECUTE ON PROCEDURE AddWardRequisition TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0037 seconds.)

```
GRANT EXECUTE ON PROCEDURE UpdateWardRequisition TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0037 seconds.)

```
GRANT EXECUTE ON PROCEDURE DeleteWardRequisition TO 'roque_charge_nurse'@'localhost';
```

SHOW GRANTS FOR 'roque_charge_nurse'@'localhost';

Grants for roque_charge_nurse@localhost

```
GRANT USAGE ON *.* TO 'roque_charge_nurse'@'localh...
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`inp...
GRANT SELECT, INSERT, UPDATE, DELETE ON `wellmeado...
GRANT SELECT ON `wellmeadows`.`staff` TO 'roque_ch...
GRANT SELECT, INSERT, UPDATE, DELETE ON `wellmeado...
GRANT SELECT, INSERT, UPDATE ON `wellmeadows`.`out...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addmedic...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwaiti...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`deleteme...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updatewa...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`deletewa...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getpatie...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getoutpa...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updateme...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwards...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addwardr...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`manageou...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`updatein...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`getwardp...
GRANT EXECUTE ON PROCEDURE `wellmeadows`.`addinpat...
```

-- Insert a New Ward Requisition (Manual requisitionID)

CALL AddWardRequisition(11, 22, 5, 5, '2025-03-15', NULL);

-- Verify that the record was inserted correctly

SELECT * FROM wardRequisition WHERE requisitionID = 11;

-- Update the record (Ensure `requisitionID = 11` exists)

CALL UpdateWardRequisition(11, 5, '2025-03-18');

-- Verify the update

SELECT * FROM wardRequisition WHERE requisitionID = 11;

-- Delete the record (Ensure `requisitionID = 11` exists before deleting)

CALL DeleteWardRequisition(11);

-- Verify that the record was deleted

SELECT * FROM wardRequisition WHERE requisitionID = 11;

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0572 seconds.)

```
-- Insert a New Ward Requisition (Manual requisitionID) CALL AddWardRequisition(11, 22, 5, 5, '2025-03-15', NULL);
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 0 (1 total, Query took 0.0008 seconds.)

```
-- Verify that the record was inserted correctly SELECT * FROM wardRequisition WHERE requisitionID = 11;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	requisitionID	staffIDPlacingReq	wardID	receivedBy	dateOrdered	dateReceived
☐ Edit <input type="image"/> Copy <input type="image"/> Delete	11	22	5	5	2025-03-15	NULL
⬅️ ☐ Check all With selected: <input type="image"/> Edit <input type="image"/> Copy <input type="image"/> Delete <input type="image"/> Export						

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0084 seconds.)

```
-- Update the record (Ensure `requisitionID = 11` exists) CALL UpdateWardRequisition(11, 5, '2025-03-18');
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

```
-- Verify the update SELECT * FROM wardRequisition WHERE requisitionID = 11;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	requisitionID	staffIDPlacingReq	wardID	receivedBy	dateOrdered	dateReceived
☐ Edit <input type="image"/> Copy <input type="image"/> Delete	11	22	5	5	2025-03-15	2025-03-18
⬅️ ☐ Check all With selected: <input type="image"/> Edit <input type="image"/> Copy <input type="image"/> Delete <input type="image"/> Export						

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0097 seconds.)

```
-- Delete the record (Ensure `requisitionID = 11` exists before deleting) CALL DeletewardRequisition(11);
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0095 seconds.)

```
-- Verify that the record was deleted SELECT * FROM wardRequisition WHERE requisitionID = 11;
```

[Edit inline] [Edit] [Create PHP code]

requisitionID staffIDPlacingReq wardID receivedBy dateOrdered dateReceived

Query results operations

-- (n) Produce a report listing the details of supplies provided to specific wards (Charge Nurse and Medical Director)

DELIMITER \$\$

-- Procedure to Get Supply Report for All Wards

CREATE PROCEDURE GetSupplyReport()

BEGIN

```
SELECT wr.requisitionID, wr.dateOrdered, wr.dateReceived,
       w.wardID, w.wardName, w.location,
       sri.itemID, sns.supplyName, sri.drugID, ps.drugName,
       sri.quantityRequired, sri.costPerUnit
```

```
FROM wardRequisition wr
```

```
JOIN ward w ON wr.wardID = w.wardID
```

```
JOIN wardRequisitionItem sri ON wr.requisitionID = sri.requisitionID
```

```
LEFT JOIN surgNonSurgSupply sns ON sri.itemID = sns.itemID
```

```
LEFT JOIN pharmaSupply ps ON sri.drugID = ps.drugID;
```

END \$\$

DELIMITER ;

GRANT EXECUTE ON PROCEDURE GetSupplyReport TO 'roque_charge_nurse'@'localhost';

GRANT EXECUTE ON PROCEDURE GetSupplyReport TO

'roque_medical_director'@'localhost';

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0171 seconds.)

```
-- Procedure to Get Supply Report for All Wards CREATE PROCEDURE GetSupplyReport() BEGIN SELECT wr.requisitionID, wr.dateOrdered, wr.dateReceived, w.wardID, w.wardName, w.location,
sri.itemID, sns.supplyName, sri.drugID, ps.drugName, sri.quantityRequired, sri.costPerUnit FROM wardRequisition wr JOIN ward w ON wr.wardID = w.wardID JOIN wardRequisitionItem sri ON
wr.requisitionID = sri.requisitionID LEFT JOIN surgNonSurgSupply sns ON sri.itemID = sns.itemID LEFT JOIN pharmaSupply ps ON sri.drugID = ps.drugID; END;
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0039 seconds.)

```
GRANT EXECUTE ON PROCEDURE GetSupplyReport TO 'roque_charge_nurse'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0051 seconds.)

```
GRANT EXECUTE ON PROCEDURE GetSupplyReport TO 'roque_medical_director'@'localhost';
```

[Edit inline] [Edit] [Create PHP code]

```

Grants for roque_charge_nurse@localhost
GRANT USAGE ON *.* TO 'roque_charge_nurse'@'localhost'
GRANT SELECT, INSERT, UPDATE ON 'wellmeadows' .inp...
GRANT SELECT, INSERT, UPDATE, DELETE ON 'wellmeado...
GRANT SELECT ON 'wellmeadows' .staff TO 'roque_ch...
GRANT SELECT, INSERT, UPDATE, DELETE ON 'wellmeado...
GRANT SELECT, INSERT, UPDATE ON 'wellmeadows' .out...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .addmedic...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .getwaiti...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .deleteeme...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .updateawa...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .deletewa...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .getpatie...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .getsuppl...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .updateme...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .getwards...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .addwardr...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .manageou...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .updatein...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .addinpat...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .getoutpa...
GRANT EXECUTE ON PROCEDURE 'wellmeadows' .getwardp...

```

```

Grants for roque_medical_director@localhost
GRANT USAGE ON *.* TO 'roque_medical_director'@'lo...
GRANT SELECT, INSERT, UPDATE, DELETE ON 'wellmeado...
GRANT SELECT ON 'wellmeadows`.`outpatient` TO 'roq...
GRANT SELECT ON 'wellmeadows`.`inpatient` TO 'roqu...
GRANT EXECUTE ON PROCEDURE 'wellmeadows`.`updatesu...
GRANT EXECUTE ON PROCEDURE 'wellmeadows`.`getoutpa...
GRANT EXECUTE ON PROCEDURE 'wellmeadows`.`deletesu...
GRANT EXECUTE ON PROCEDURE 'wellmeadows`.`getwaiti...
GRANT EXECUTE ON PROCEDURE 'wellmeadows`.`addsuppl...
GRANT EXECUTE ON PROCEDURE 'wellmeadows`.`getwardp...
GRANT EXECUTE ON PROCEDURE 'wellmeadows`.`getsuppl...

```

✔ Showing rows 0 - 9 (10 total, Query took 0.0031 seconds.)

[CALL](#) `getSupplyReport();`

[\[Edit inline\]](#) [\[Edit\]](#) [\[Create PHP code\]](#)

☐ Show all | Number of rows: Filter rows:

[Extra options](#)

requestionID	dateOrdered	dateReceived	wardID	wardName	location	itemID	supplyName	drugID	drugName	quantityRequired	costPerUnit
1	2025-03-01	2025-03-02	3	Neurology	C Block	1	Sterile Gloves	NULL	NULL	50	2.50
2	2025-03-03	2025-03-04	4	Pediatrics	B Block	2	Surgical Mask	NULL	NULL	100	1.25
3	2025-03-05	2025-03-06	5	Oncology	A Block	3	Scalpel Blades	NULL	NULL	200	5.00
4	2025-03-07	2025-03-08	6	Geriatrics	F Block	4	Syringe 10ml	NULL	NULL	150	0.75
5	2025-03-09	2025-03-10	7	Emergency	G Block	5	Bandages	NULL	NULL	300	3.00
6	2025-03-11	2025-03-12	8	ICU	H Block	NULL	NULL	1	Paracetamol	500	0.50
7	2025-03-13	2025-03-14	9	Surgery	I Block	NULL	NULL	2	Amoxicillin	200	1.00
8	2025-03-15	2025-03-16	10	Psychiatry	J Block	NULL	NULL	3	Ibuprofen	150	0.75
9	2025-03-17	2025-03-18	11	Maternity	K Block	NULL	NULL	4	Omeprazole	100	1.50
10	2025-03-19	2025-03-20	12	Nephrology	L Block	NULL	NULL	5	Metformin	250	2.00